

Hoon Lee

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Research Interest

I'm interested in developing a scalable reinforcement learning framework for humanoid robots that can continually adapt and generalize. Previous works include:

- Scalable RL & Plasticity ([Preprint'26](#), [Preprint'26](#), [ICLR'26](#), [ICML'25](#), [ICLR'25](#), [ICML'24](#), [NeurIPS'23](#)).
- Robotics ([Preprint'26](#), [Preprint'26](#), [ICRA'26](#), [RA-L'25](#)).
- RL Post-training ([COLMw'25](#), [WWW'22](#)).
- Self-supervised learning ([ICML'24](#), [ICML'23](#)).
- Skill Discovery ([NeurIPS'24](#), [NeurIPS'23](#)).

Education

2022.03 – 2026.02 **KAIST, PhD** in Artificial Intelligence (Advisor: [Jaegul Choo](#)).
Thesis: *Maintaining Plasticity for Scalable Deep Reinforcement Learning*.
Committee: [Jaegul Choo](#), [Chulhee Yun](#), [Kimin Lee](#), [Clare Lyle](#), [Peter Stone](#).
[slide](#)

2020.03 – 2022.02 **KAIST, MS** in Artificial Intelligence.
Thesis: *Personalized Draft Recommendation System for Victory in League of Legends*.
[thesis](#)

2014.03 – 2020.02 **Korea University, BS** in Computer Science.

Work Experience

2026.01 – Current **Senior Research Scientist @ Holiday Robotics**
Designed an autonomous reward design agent for dexterous manipulation.

2025.05 – 2025.11 **Research Intern @ Meta Reality Labs** (Mentor: [Nitin Kamra](#), [Karl Ridgeway](#))
Designed an autonomous reward design agent for dexterous manipulation.
[paper](#)

2025.02 – 2025.04 **Research Intern @ Krafton AI** (Mentor: [Dongmin Park](#), [Jongho Park](#))
Developed a LLM-based chess-playing agent by RLVR.
[paper](#)

2024.02 – 2024.08 **Research Intern @ Sony AI** (Mentor: [Takuma Seno](#), [Kaushik Subramanian](#), [Peter Stone](#))
Developed a vision-based autonomous racing agent in GranTurismo 7.
[paper](#), [video](#)

2021.09 – 2021.11 **Research Intern @ Kakao Enterprise** (Mentor: [Kyushik Min](#))
Implemented an open-source reinforcement learning library for research.
[code](#) (300+ ☆)

2019.03 – 2019.07 **Research Intern @ Neowiz** (Mentor: [Jaejin Yoon](#))
Developed a reinforcement learning agent for the tactical RPG game, Browndust.
[poster](#)

Research Topics and Publications

Maintaining Plasticity for Scalable RL

- Preprint'26* **Unleashing the Architectural Potential of RL in Visual Continuous Control**
*Donghu Kim, Youngdo Lee, **Hojoon Lee**, Johan Obando-Ceron, ByungKun Lee, Aaron Courville, Pablo Samuel Castro, Jaegul Choo, Clare Lyle.*
[paper](#)
- Preprint'26* **FlashSAC: Fast and Stable Off-Policy RL for High-Dimensional Robot Control**
Donghu Kim, Youngdo Lee*, Minho Park, Kinam Kim, Takuma Seno, I Made Aswin Nahendra, Sehee Min, Daniel Palnicek, Florian Vogt, Danica Kraig, Jan Peters, Jaegul Choo[†], **Hojoon Lee**[†].*
[paper](#), [website](#), [code](#)
- ICLR'26*
(oral) **FIRE: Frobenius-Isometry Reinitialization for Balancing Stability-Plasticity Tradeoff**
*Isaac Han, Sangyeon Park, Seungwon Oh, Donghu Kim, **Hojoon Lee**, Kyungjoon Kim.*
[paper](#), [website](#)
- ICML'25*
(spotlight) **SimbaV2: Hyperspherical Normalization for Scalable RL**
Hojoon Lee, Youngdo Lee*, Takuma Seno, Donghu Kim, Peter Stone, Jaegul Choo.*
[paper](#), [website](#), [code](#) (100+☆)
- ICLR'25*
(spotlight) **Simba: Simplicity Bias for Scaling Up Parameters in Deep RL**
Hojoon Lee, Dongyoon Hwang*, Donghu Kim, Hyunseung Kim, Jun Jet Tai, Kaushik Subramanian, Peter R. Wurman, Jaegul Choo, Peter Stone, Takuma Seno.*
[paper](#), [website](#), [code](#) (100+☆)
- ICML'24* **Slow and Steady Wins the Race: Maintaining Plasticity with Hare and Tortoise**
***Hojoon Lee**, Hyeonseo Cho, Hyunseung Kim, Donghu Kim, Jaegul Choo, Clare Lyle.*
[paper](#)
- NeurIPS'23* **PLASTIC: Improving Input and Label Plasticity for Sample-Efficient RL**
Hojoon Lee, Hanseul Cho*, Hyunseung Kim*, Daehoon Gwak, Joonkee Kim, Jaegul Choo, Seyoung Yun, Chulhee Yun.*
[paper](#)

Robotics

- Preprint'26* **3D HAMSTER: Hierarchical VLAs through 3D Trajectory Guidance**
Dongyoon Hwang, ByungKun Lee*, Dongjin Kim, Hyojin Jang, Hoiyeong Jin, Jueun Mun, Minho Park, **Hojoon Lee**, Hyunseung Kim, Jaegul Choo.*
coming soon
- Preprint'26* **RDA: Reward Design Agent for Reinforcement Learning**
***Hojoon Lee**, Ajay Subramanian, Ben Abbatematteo, Pedro Matias, Vijay Veerabadran, Karl Ridgeway, Nitin Kamra.*
[paper](#)

Preprint'26 **PHUMA: Physically Grounded Humanoid Locomotion Dataset**
Kyungmin Lee*, Sibeon Kim*, Minho Park, Dongyoon Hwang, **Hojoon Lee**[†], Jaegul Choo[†].
[paper](#), [website](#), [code](#) (150+☆)

ICRA'26 **ACG: Action Coherence Guidance For Flow-based VLA Models.**
Minho Park*, Kinam Kim*, Junha Hyung, Hyojin Jang, Hoiyeong Jin, Jooyeol Yun,
Hojoon Lee[†], Jaegul Choo[†].
[paper](#), [website](#), [code](#)

RA-L'25 **A Champion-level Vision-based Racing Agent for Competitive Racing in GT7.**
Hojoon Lee, Takuma Seno, Jun Jet Tai, Kaushik Subramanian, Kenta Kawamoto,
Peter Stone, Peter R. Wurman.
[paper](#), [video](#)

RL Post-training

CoLMw'25 **Can Large Language Models Develop Strategic Reasoning? Post-training insights from Learning to Play Chess**
Dongyoon Hwang*, **Hojoon Lee***, Jaegul Choo, Dongmin Park, Jongho Park
[paper](#)

WWW'22 **DraftRec: Personalized Draft Recommendation System for Victory in MOBA games**
Hojoon Lee*, Dongyoon Hwang*, Hyunseung Kim, Byungkun Lee, Jaegul Choo
[paper](#)

Self-Supervised Learning

ICML'24 **Investigating Pre-training Objectives for Generalization in Vision-Based RL**
Donghu Kim*, **Hojoon Lee***, Kyungmin Lee*, Dongyoon Hwang, Jaegul Choo.
[paper](#)

ICML'23 **On the importance of Feature Decorrelation for Representation Learning in RL**
Hojoon Lee, Gwanho Lee, Dongyoon Hwang, Hyunho Lee, Byungkun Lee, Jaegul Choo.
[paper](#)

Skill Discovery

NeurIPS'24 **Do's and Don'ts: Learning Desirable Skills with Instruction Videos**
Hyunseung Kim, Byungkun Lee, **Hojoon Lee**, Dongyoon Hwang, Donghu Kim, Jaegul Choo.
[paper](#)

NeurIPS'23 **Learning to Discover Skills through Guidance**
Hyunseung Kim*, Byungkun Lee*, **Hojoon Lee**, Dongyoon Hwang, Kyuhsik Min, Jaegul Choo.
[paper](#)

Mentoring

2025.03 – 2025.10 [Isaac Han](#): Advised on [ICLR'26](#) (GIST PhD)
2025.03 – 2025.10 [Kyungmin Lee](#): Advised on [Preprint'25](#) (KAIST PhD)
2025.03 – 2025.10 [Sibeen Kim](#): Advised on [Preprint'25](#) (KAIST MS)
2025.03 – 2025.10 [Minho Park](#): Advised on [ICRA'26](#) (KAIST PhD)
2024.10 – 2025.03 [Donghu Kim](#): Mentored on [Preprint'25](#), [ICML'24](#) (KAIST MS → Holiday Robotics)
2024.10 – 2025.03 [Youngdo Lee](#): Mentored on [ICML'24](#) (KAIST MS → Krafton AI)
2023.09 – 2024.02 [Hyeonseo Cho](#): Mentored on [ICML'24](#) (Konkuk BS → KAIST MS)
2021.09 – 2023.02 [Dongyoon Hwang](#): Co-authored 5 papers (Korea Univ BS → KAIST PhD)
2021.09 – 2023.02 [Hyunseung Kim](#): Co-authored 4 papers (Korea Univ BS → KAIST PhD → Krafton AI)

Talks and Presentations

2024.11 [BeNeRL](#): Designing Neural Network Architecture for Deep RL [slide](#)
2024.05 [Sony AI](#): Towards Plastic Neural Network [slide](#)
2024.02 [Konkuk University](#): Towards Plastic Neural Network [slide](#)
2023.12 [RL Korea](#): Pretraining for Intelligent Reinforcement Learning Agent [slide](#)

Honors and Awards

2023.06 [Crevisse Partners](#): CIKM Travel Award (3,000 USD)
2022.10 [ACM SIGIR](#): Best Short Paper Honorable Mention Award
2021.03 [Korea Government](#): Full Academic Scholarship (10,000 USD)
2019.12 [Korea University](#): Graduation Project Award (1,500 USD)
2017.09 [Seongnam City](#): College Scholarship (4,000 USD)
2017.02 [US Army](#): General Paik Sun Yup Leadership Award [article](#)

Academic Services

Reviewer [NeurIPS](#), [ICML](#), [ICLR](#), [AAAI](#), [CoLLAs](#), [ICRA](#), [IROS](#), [CoG](#)

Skills

Technical [C](#), [Python](#), [PyTorch](#), [Jax](#), [Tensorflow](#), [Git](#), [CI/CD](#), [Docker](#), [Isaac](#), [Mujoco](#)
Lingual [Korean](#) (native), [English](#) (business fluent)

Military Services

2015.04-2017.03 [KATUSA](#) (Korean Augmentation to the U.S. Army), Discharged as Sergeant